

DAA Assignment – Set 7

Question 1 – Application Using Heap (K Largest Elements / Priority Queue)

Given an array $A[0 \dots n-1]$ of integers and an integer k , design an algorithm using a heap (priority queue) to find the k largest elements in the array.

Your solution should be efficient compared to sorting the entire array.

Tasks

1. Design an algorithm using:
 - Min-heap of size k (preferred approach)
2. Implement the algorithm.
3. Explain how heap operations help reduce time complexity.
4. Derive the time complexity of the algorithm.
5. Compare with:
 - Sorting-based approach ($\Theta(n \log n)$)
 - Brute-force selection method
6. Experiment with different values of n and k and analyze performance.

Question 2 – String Matching (Knuth–Morris–Pratt – KMP Algorithm)

Given a text string T of length n and a pattern P of length m , design an efficient algorithm to find all occurrences of P in T using the Knuth–Morris–Pratt (KMP) algorithm.

The algorithm avoids redundant comparisons using a preprocessing step.

Tasks

1. Explain the idea of the Longest Prefix Suffix (LPS) array.
2. Construct the LPS array for a given pattern.
3. Design the KMP algorithm using the LPS array.
4. Implement the algorithm.
5. Derive the time complexity of:
 - LPS construction
 - Pattern matching phase
6. Compare experimentally with:
 - Brute-force string matching

Question 3 – Divide and Conquer (Counting Inversions in an Array)

You are given an array $A[0 \dots n-1]$ of integers.

An inversion is a pair of indices (i, j) such that:

- $i < j$ and
- $A[i] > A[j]$

Design a divide-and-conquer algorithm to count the total number of inversions in the array.

Tasks

1. Design the divide-and-conquer algorithm based on merge sort.
2. Explain how inversions are counted during the merge step.
3. Implement the algorithm in a programming language of your choice.
4. Derive the recurrence relation for the running time.
5. Solve the recurrence and determine the asymptotic complexity.
6. Compare experimentally with the naïve $\Theta(n^2)$ approach.